



BUSINESS REQUIREMENTS DOCUMENT

NAVPRO — Kajian Kelayakan Finansial

Integrasi BRD KKF Proyek Investasi v1.0 (ICON+) dengan Revisi Arsitektur NAVPRO v2.0

DOKUMEN ID

BRD-KKF-NAV-002

VERSI

2.1 — Final

TANGGAL

28 Mei 2026

ENTITAS

ICON+ / PLN Group

KLASIFIKASI

Internal Confidential

STATUS

Draft for Review

1 Executive Summary



Dokumen ini menyelaraskan **BRD KKF Proyek Investasi v1.0** (Excel 1-tahun, ICON+) dengan **Revisi NAVPRO v2.0** — mencakup struktur organisasi 4 Pusat + 9 SBU, alur approval Asman→Manager, dan dukungan durasi 1–120 bulan.

KKF (Kajian Kelayakan Finansial) adalah gerbang wajib sebelum ICON+ menyetujui CAPEX last-mile fiber untuk setiap proyek investasi. Platform NAVPRO mendigitalisasi seluruh proses ini — menggantikan template Excel terisolasi menjadi sistem enterprise terintegrasi.

REDUKSI WAKTU KKF

~90%

3–5 hari → <4 jam

REDUKSI WAKTU
APPROVAL

~80%

2–5 hari → <24 jam

KONSISTENSI ASUMSI

100%

dari ~60% (manual)

AUDIT TRAIL
COVERAGE

100%

dari 0% (Excel)

Evolusi BRD v1.0 → NAVPRO v2.0

Aspek

BRD v1.0 (docx)

NAVPRO v2.0 (implementasi)

Durasi proyek

Kontrak ≤ 12 bulan

1–120 bulan + preset
SHORT/MID/LONG/EXTENDED

Alur approval

SA → Manager → GM/SRM

Staff(L4) → Asman(L3) → Manager
Segment(L2)

Struktur
organisasi

Tidak dimodelkan per SBU

4 Pusat + 9 SBU regional, org_unit_id,
segment

Project ID	KKF-[YEAR]-[SEQ]	NAVPRO-YYYY-####
Kesimpulan	LAYAK / TIDAK LAYAK / BUTUH JUSTIFIKASI	LAYAK / BERSYARAT / TIDAK_LAYAK / WASPADA
Tech stack	Next.js + API (generik)	Node.js + Express + PostgreSQL + Static HTML/JS

Parameter Governance Keuangan

Parameter	Nilai	Sumber
WACC Annual	9,72% p.a.	Memo VP Keuangan April 2026
BCR Mandatory	≥ 1,23	Memo DirKeu 16 Sep 2025
BCR Minimum	≥ 1,08	Memo DirKeu 16 Sep 2025
Inflasi Bulanan Default	0,3% per bulan	Assumption Master
Durasi Proyek	1–120 bulan	System Config
Formula XIRR	Newton-Raphson (max 1000 iter, tol 1e-7)	Calculation Engine

2

Latar Belakang & Problem Statement

2.1 Profil Organisasi

ICON+ (PT Indonesia Comnets Plus) adalah anak perusahaan telekomunikasi PLN Group yang menyediakan layanan konektivitas termasuk solusi fiber optik last-mile untuk klien enterprise dan pemerintah di seluruh Indonesia. Perusahaan mengelola deployment multi-site meliputi Sumatera, Jawa, Kalimantan, Sulawesi, dan Indonesia Timur.

2.2 Masalah Utama

ID	Masalah	Dampak Bisnis	Prioritas
P-01	File Excel tersebar, tidak terpusat	Keputusan investasi berbasis data salah; version chaos	Critical
P-02	Asumsi WACC/BCR tidak konsisten	Pelanggaran kebijakan DirKeu; inkonsistensi antar file	Critical
P-03	Tidak ada audit trail	Non-compliant PLN Group governance; formula corruption	Critical
P-04	Approval manual via kertas/email	SLA tidak terukur, tidak ada eskalasi otomatis	High

P-05	Zero visibilitas eksekutif	Manajemen tidak dapat memantau pipeline proyek	High
P-06	Duplikasi data lintas sistem	Overhead rekonsiliasi SAP/iCRM/LKPP/Excel	Medium
P-07	Skalabilitas terbatas (1 proyek/file)	Analisis portofolio tidak mungkin tanpa konsolidasi manual	High

2.3 Proses AS-IS (7 Langkah Excel)

Langkah	Sheet Excel v01	Aktivitas	Modul NAVPRO
1	02_Inv_Details	RAB Lastmile — 8 item investasi standar	Wizard RAB/CAPEX
2	03_CAPEX	Mapping CAPEX — 4 kategori + PM + Other	Wizard CAPEX
3	04_OPEX_Bulanan	OPEX bulanan — koefisien + inflasi otomatis	Wizard OPEX
4	05_Pendapatan	Revenue — hingga 12 service lines	Wizard Revenue
5	06_Asumsi	Review asumsi WACC, BCR, inflasi, FX	Assumption Master
6	07_Cashflow_Bulanan	XIRR, XNPV, BCR, Payback — auto-kalkulasi	Calculation Engine
7	08_Ex_Summary	Executive Summary + KPI status + tanda tangan	ExecutiveSummary + PDF

2.4 Formula Kritis AS-IS

Formula	Logika Excel	Engine NAVPRO
OPEX Bulanan	$\text{Baseline} \times (1 + \text{inflasi_bulanan})^{(\text{bulan}-1)}$	computeCashflowMonthly
Active Flag	$\text{Bulan} > \text{durasi kontrak} \rightarrow 0$	Logika identik
XNPV	Discount exact-date, WACC tahunan	calculateXNPV
XIRR	Newton-Raphson, max 1000 iter, tol 1e-7	calculateXIRR
BCR/PI	$\text{PV inflow } M1-Mn \div \text{CAPEX } M0 $	computeProjectKpi
Payback	Bulan pertama cumulative CF ≥ 0	calculatePayback
Simple ROI	$\text{Net inflow } M1-Mn \div \text{CAPEX } M0 $	Tersimpan di KPI

Auto-
conclusion

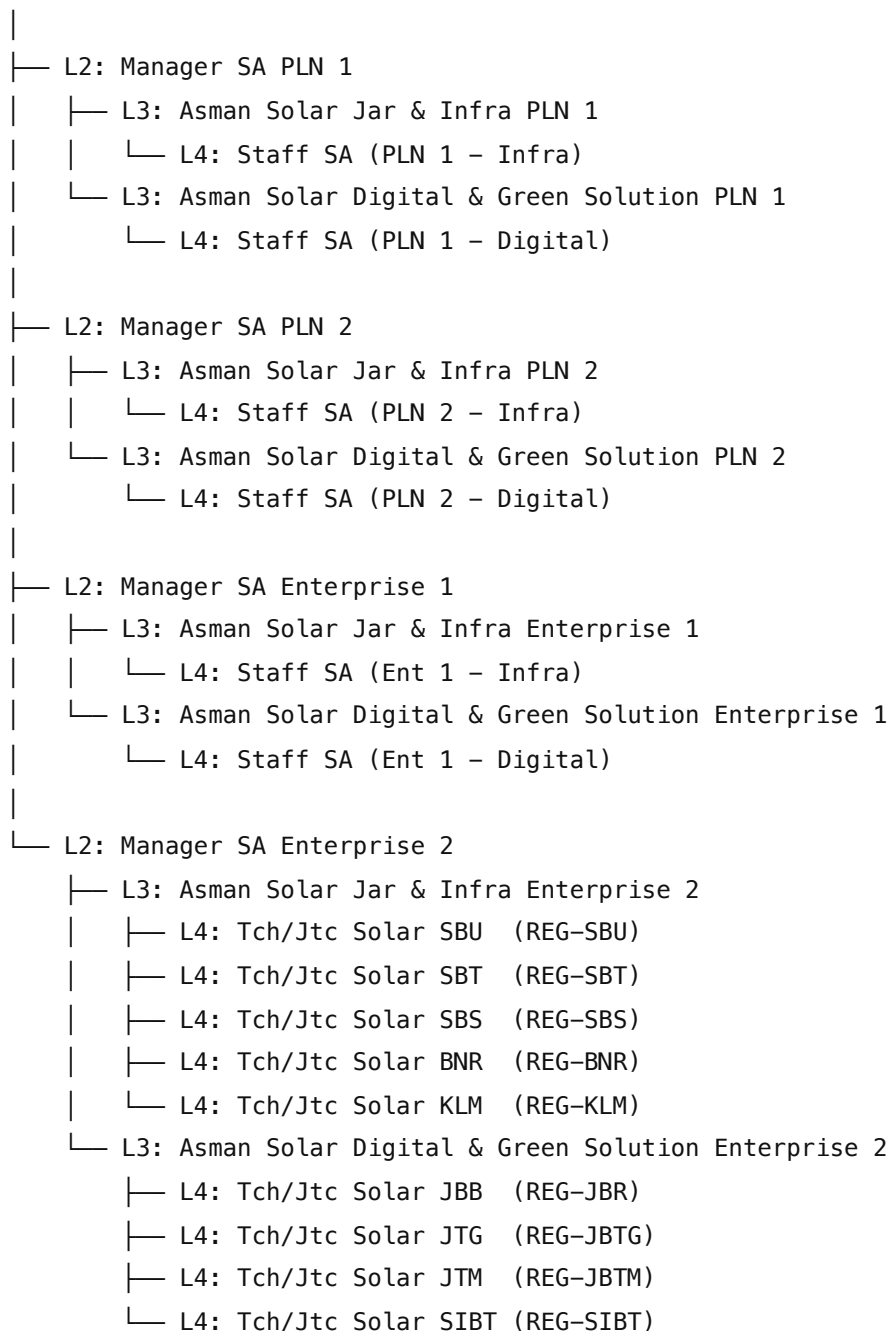
$NPV > 0 \wedge XIRR \geq WACC \wedge BCR \geq \text{ambang} \wedge$
payback < durasi

LAYAK/BERSYARAT/TIDAK_LAYAK

3 Struktur Organisasi

3.1 Hierarki 4 Level

L1: VP Solution Architect



Apabila ada request untuk PLN, Solar SBU akan disupervisi oleh Bidang Solar PLN.

3.2 Unit Pusat (4 Sub-Bidang)

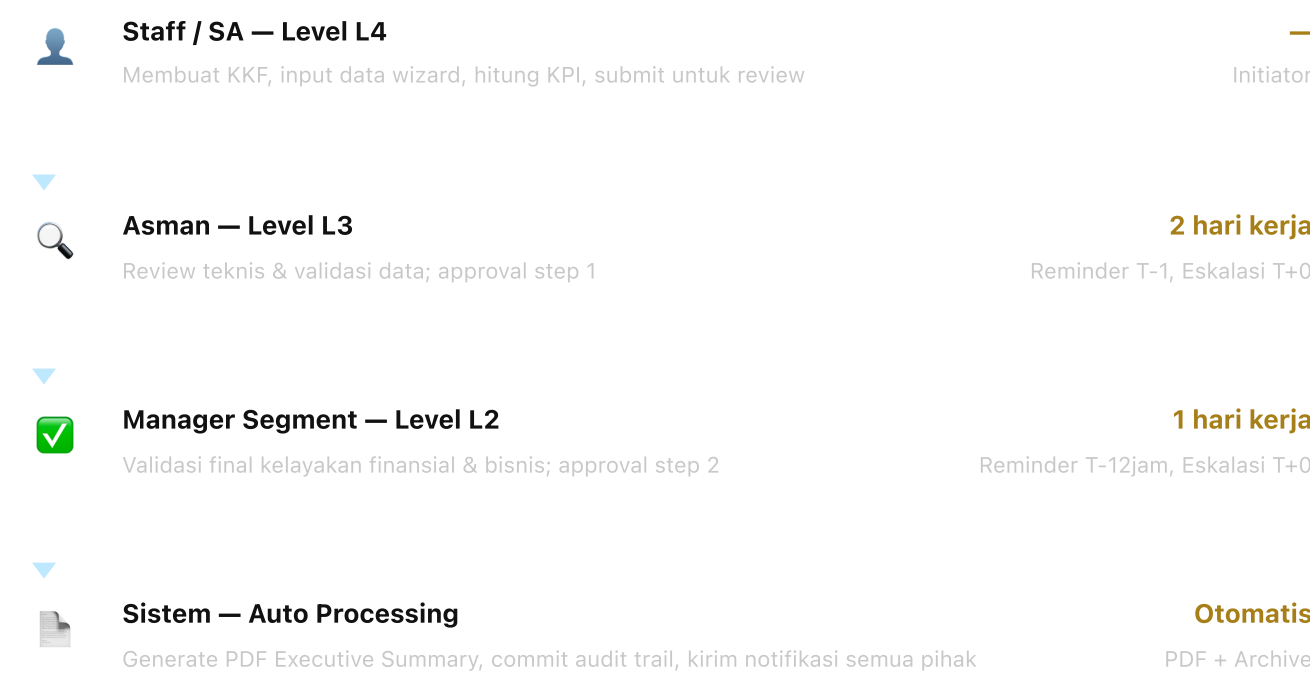
3.3 SBU Regional (9 Unit)

Kode	Nama	Segment	Kode	Nama	Segment
SOLAR-ENT-1	Sub Bid SA Enterprise 1	ENT1	REG-SBU	Tch/Jtc Solar SBU	ENT2
			REG-SBT	Tch/Jtc Solar SBT	ENT2
SOLAR-ENT-2	Sub Bid SA Enterprise 2	ENT2	REG-SBS	Tch/Jtc Solar SBS	ENT2
			REG-JBR	Tch/Jtc Solar JBB	ENT2
SOLAR-PLN-1	Sub Bid SA PLN 1	PLN1	REG-JBTG	Tch/Jtc Solar JTG	ENT2
			REG-JBTM	Tch/Jtc Solar JTM	ENT2
SOLAR-PLN-2	Sub Bid SA PLN 2	PLN2	REG-BNR	Tch/Jtc Solar BNR	ENT2
			REG-KLM	Tch/Jtc Solar KLM	ENT2
			REG-SIBT	Tch/Jtc Solar SIBT	ENT2

4

Hierarki Approval & Business Flow

4.1 Rantai Approval



2 Input Data — Wizard 6 Langkah (Staff L4)

RAB Lastmile (8 item) → CAPEX Mapping → OPEX Bulanan (7 komponen, inflasi otomatis) → Revenue Composition (max 12 service lines) → Review Asumsi (read-only) → Auto-save setiap 2 menit

3 Kalkulasi Finansial (Sistem)

Staff trigger "Hitung". Engine: XIRR, XNPV, BCR/PI, Payback, Simple ROI. Auto-conclusion: LAYAK / BERSYARAT / TIDAK LAYAK. Status: COMPUTED

4 Submit untuk Review (Staff L4)

Staff review hasil kalkulasi, klik Submit. Sistem: lock data, buat snapshot calculation_versions, kirim notifikasi ke Asman terkait. Status: SUBMITTED → IN_REVIEW_ASMAN

5 Review Asman (Asman L3) — SLA 2 hari kerja

Asman review data teknis & KPI. [APPROVE] → Status: IN_REVIEW_MANAGER, notifikasi Manager. [REJECT] → Status: DRAFT, catatan wajib ≥20 karakter, notifikasi Staff dengan alasan penolakan

6 Approval Final Manager (Manager L2) — SLA 1 hari kerja

Manager validasi kelayakan finansial & bisnis. [APPROVE] → Status: APPROVED, generate PDF, notifikasi semua pihak. [REJECT] → Status: DRAFT, notifikasi Staff & Asman

7 Selesai (Sistem)

PDF Executive Summary auto-generated, audit trail final di-commit, notifikasi ke seluruh stakeholder, proyek diarsipkan. Status: APPROVED/ARCHIVED

4.3 Business Rules Kritis**Financial Rules**

ID	Rule
BR-F01	LAYAK: NPV>0 AND XIRR>WACC AND BCR≥1.23 AND Payback<Durasi
BR-F02	BERSYARAT: BCR≥1.08 (dengan justifikasi wajib)
BR-F03	TIDAK LAYAK: BCR<1.08
BR-F04	WACC hanya dapat diubah Finance Admin + referensi memo wajib
BR-F05	BCR threshold: perubahan wajib referensi memo DirKau

Approval Rules

ID	Rule
BR-A01	KKF hanya dapat di-submit jika semua field mandatory terisi
BR-A02	Data KKF terkunci setelah status SUBMITTED
BR-A03	Rejection wajib disertai catatan minimal 20 karakter
BR-A04	SLA Asman: 2 hari kerja sejak submission
BR-A05	SLA Manager: 1 hari kerja sejak approval Asman
BR-A06	Approval Manager valid hanya jika Asman segment sama sudah approve

5 Core Features (F-01 – F-12)

ID	Fitur	Deskripsi	Status
F-01	Authentication & Session	Login JWT, MFA untuk Manager/GM, logout 5x gagal, session 8 jam	✓ Done
F-02	User Management & RBAC	7 role: SUPER_ADMIN, FINANCE_ADMIN, VP_SA, MANAGER, ASMAN, STAFF, VIEWER	✓ Done
F-03	Assumption Master	WACC, BCR thresholds, inflasi, FX — terpusat dengan riwayat perubahan	✓ Done
F-04	KKF Lifecycle Management	Status machine DRAFT→APPROVED, versioning, clone, soft-delete	✓ Done
F-05	Financial Data Entry Wizard	7 modul: RAB, CAPEX, OPEX, Revenue, Asumsi, Cashflow Preview, Exsum Preview	✓ Done
F-06	Calculation Engine	Server-side: XIRR, XNPV, BCR/PI, Payback, Simple ROI, auto-conclusion	✓ Done
F-07	Approval Workflow Engine	Multi-level: Asman(SLA 2hr) → Manager(SLA 1hr); delegation; SLA reminder	✓ Done
F-08	Executive & Operational Dashboard	Portfolio KPI, approval pipeline, heatmap, per-segment analytics	✓ Done
F-09	Notification System	In-app + email untuk semua workflow events dan SLA reminders	⌂ Partial
F-10	Audit Trail	Immutable log setiap aksi: field-level changes, approval, login, export	✓ Done
F-11	Excel Import/Export	Import template KKF v01; export PDF Executive Summary & Excel cashflow	⌂ Partial
F-12	Reporting Module	Portfolio report, compliance report, scheduled delivery; PDF & Excel	⌚ Phase 2

6 Functional Requirements

ID	Modul	Kebutuhan
FR-001	Auth	Autentikasi via username dan password

FR-002	Auth	MFA untuk Manager dan GM/SRM
FR-003	Auth	Lockout akun setelah 5 gagal login selama 15 menit
FR-010	User Mgmt	Admin dapat create/edit/deactivate user
FR-011	User Mgmt	Assignment role + org_unit_id + org_level per user
FR-012	User Mgmt	WACC dan BCR threshold hanya dapat diubah Finance Admin
FR-020	Assumption	Finance Admin update asumsi dengan referensi sumber wajib
FR-021	Assumption	Riwayat perubahan: nilai lama, baru, user, timestamp, sumber
FR-022	Assumption	Notifikasi ke semua SA dengan KKF pending saat WACC diupdate
FR-030	Project	Generate Project ID unik (NAVPRO-YYYY-####)
FR-031	Project	Status machine: DRAFT→SUBMITTED→IN_REVIEW_ASMAN→IN_REVIEW_MANAGER→APPROVED/REJI
FR-034	Project	Data terkunci setelah status SUBMITTED; hanya bisa diubah via rejection
FR-040	RAB	8 item RAB standar sesuai template Excel
FR-050	CAPEX	Auto-link Perangkat Telekomunikasi (1b) dari total RAB
FR-060	OPEX	Kalkulasi OPEX: Baseline = Koefisien × Revenue bulanan
FR-061	OPEX	OPEX bulanan dengan inflasi compounding: $\text{Baseline} \times (1+r)^{(m-1)}$
FR-080	Calc	Net Cashflow untuk 13 periode (Month 0 s/d Month 12+)
FR-081	Calc	Active Flag: bulan > durasi kontrak → nilai = 0

FR-082	Calc	XIRR via Newton-Raphson, initial guess 10%
FR-088	Calc	Auto-conclusion: LAYAK jika NPV>0 AND XIRR>WACC AND BCR≥1.23 AND Payback<D
FR-090	Approval	Submit KKF (semua field valid) → routing ke Asman terkait
FR-091	Approval	Notifikasi Asman segera setelah Staff submit
FR-092	Approval	SLA Asman: 2 hari kerja; SLA Manager: 1 hari kerja
FR-094	Approval	Rejection wajib komentar minimal 20 karakter
FR-095	Approval	Rejection → project kembali DRAFT dengan alasan visible ke Staff
FR-096	Approval	Semua keputusan approval: timestamp + user + immutable audit trail
FR-120	Audit	Log setiap perubahan field: nama field, nilai lama, baru, user, timestamp
FR-121	Audit	Audit log immutable — tidak ada user yang dapat menghapus

7 Non-Functional Requirements

ID	Kategori	Kebutuhan	Target
NFR-P01	Performance	API response time	≤ 200ms (p95)
NFR-P02	Performance	Waktu kalkulasi semua KPI	≤ 3 detik (12 bulan), ≤ 10 detik (120 bulan)
NFR-P03	Performance	Generate PDF Executive Summary	≤ 10 detik
NFR-P04	Performance	Concurrent users	≥ 50 tanpa degradasi; peak 200 <20% degradasi
NFR-A01	Availability	Uptime bulanan	≥ 99,5% (<3,6 jam downtime/bulan)

NFR-A03	Availability	Failover komponen kritis	<30 detik
NFR-S01	Security	Password hashing	bcrypt, min cost factor 12
NFR-S02	Security	Enkripsi data transit/rest	TLS 1.2+, AES-256
NFR-S03	Security	RBAC enforcement	Server-side validation pada setiap API endpoint
NFR-S05	Security	Rate limiting API	100 req/menit per user
NFR-B01	Backup	Database backup	Daily full + hourly incremental, retensi 30 hari
NFR-B03	DR	RPO/RTO	RPO: 1 jam; RTO: 4 jam
NFR-C01	Compliance	Audit log retention	Minimum 5 tahun; approval financial: 7 tahun
NFR-C02	Compliance	Data residency	Indonesia-based DC (GovCloud / PLN-approved)
NFR-SC02	Scalability	Volume data	10.000+ KKF tanpa degradasi
NFR-UX02	Usability	Mobile responsiveness	Tablet (768px), mobile (375px); approval aksi tersedia di mobile
NFR-UX03	Usability	Bahasa & format angka	Bahasa Indonesia; locale ID (pemisah ribuan: titik, desimal: koma)

8 Role & Permission Matrix

8.1 Definisi Role

Role	Level	Scope Data	Tanggung Jawab Utama
SUPER_ADMIN	Sistem	Semua data	Administrasi penuh, user management, konfigurasi sistem
FINANCE_ADMIN	Pusat	Semua KKF (read) + Assumption (write)	Kelola asumsi WACC, BCR, inflasi, FX

VP_SA	L1	Semua bidang	Visibilitas portofolio strategis seluruh segment
MANAGER	L2	KKF segment sendiri	Approval final per segment; visibilitas portofolio segment
ASMAN	L3	KKF unit sendiri	Review teknis; approval step 1
STAFF	L4	KKF milik sendiri	Input data KKF, hitung, submit
VIEWER	—	KKF approved (read-only)	Auditor internal; review laporan

8.2 Permission Matrix Modul

Modul / Aksi	SUPER_ADMIN	FINANCE_ADMIN	VP_SA	MANAGER	ASMAN	STAFF	VIEWER
Buat KKF	✓	—	—	—	—	✓	—
Edit KKF (DRAFT)	✓	—	—	—	—	✓ (own)	—
Submit KKF	✓	—	—	—	—	✓ (own)	—
Review/Approve (Asman)	✓	—	—	—	✓ (unit)	—	—
Approve Final (Manager)	✓	—	—	✓ (segment)	—	—	—
View KKF	✓ (all)	✓ (all)	✓ (all)	✓ (segment)	✓ (unit)	✓ (own)	✓ (approved)
Assumption Master — Edit	✓	✓	—	👁	👁	👁	—
User Management	✓	—	—	—	—	—	—
Audit Log	✓	✓	👁	—	—	—	👁
Dashboard Eksekutif	✓	✓	✓	✓	—	—	—
Dashboard Operasional	✓	—	—	✓	✓	✓	—
Export PDF/Excel	✓	✓	✓	✓	✓	✓ (own)	—



9 User Flows

9.1 Flow Login

- 1 Akses URL aplikasi**
Render halaman login ICON+/PLN
- 2 Input credentials**
Email + Password → validasi server
- 3 Gagal 5x → logout 15 menit**
Error message + audit log
- 4 MFA (Manager/GM)**
OTP ke email/authenticator
- 5 Issue JWT (8 jam / 4 jam admin)**
Redirect ke halaman sesuai role

9.2 Flow Rejection & Revision

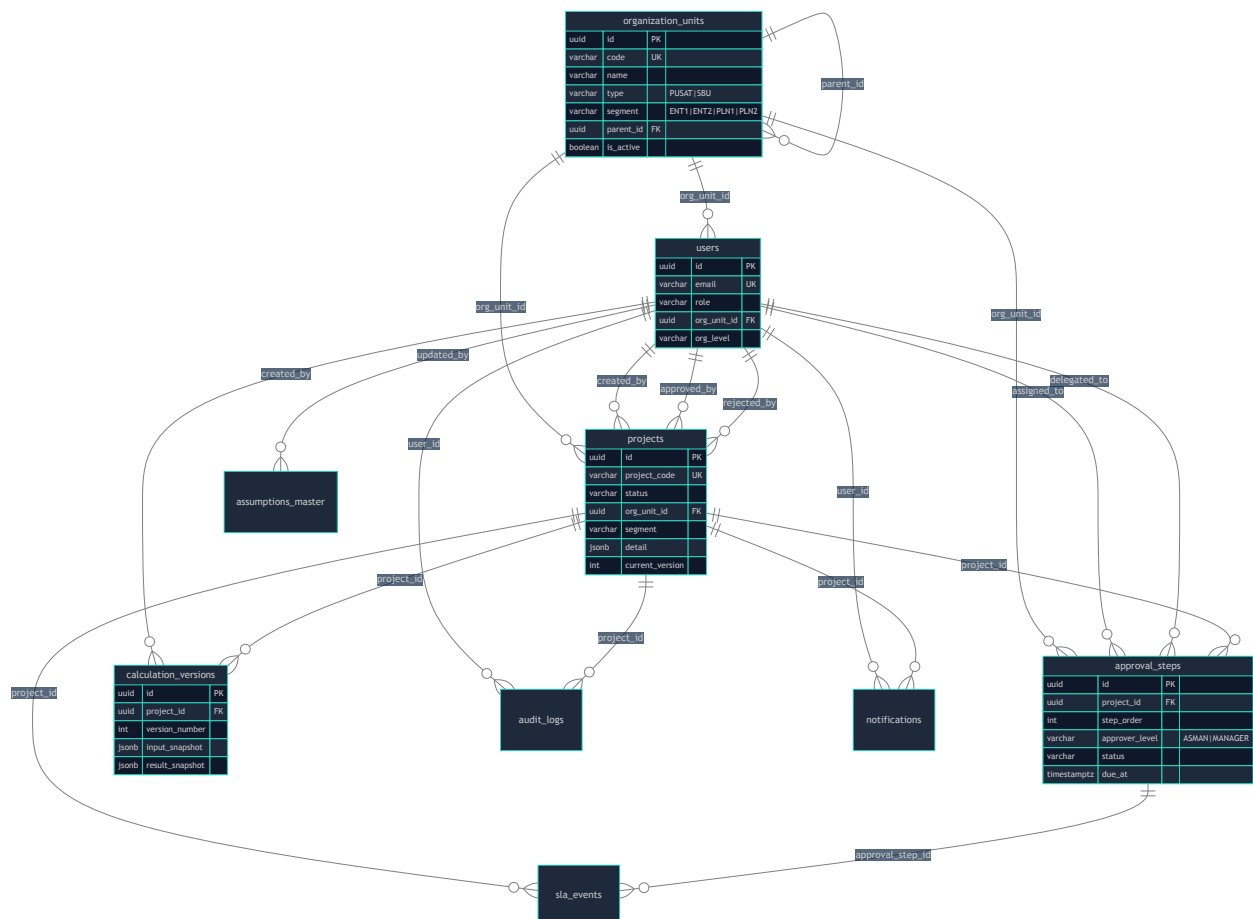
- 1 Staff terima notifikasi rejection**
Alasan penolakan tampil prominent
- 2 Status kembali ke DRAFT**
Sistem unlock project data fields
- 3 Staff revisi data**
Sistem recalculate KPI otomatis setiap perubahan
- 4 Re-submit**
Sistem buat versi baru; versi lama tersimpan di history
- 5 Approval chain restart dari step 1**
Notifikasi ulang ke Asman

9.3 Flow Eskalasi SLA

- T-1 Reminder otomatis**
Sistem kirim notifikasi ke approver aktif: "KKF [ID] membutuhkan persetujuan Anda sebelum [tanggal/jam]"
- T+0 SLA habis — Eskalasi**
Jika tidak ada aksi: sistem kirim alert eskalasi ke level atas (Asman → Manager, Manager → VP_SA). SLA breach tercatat di audit log dan dashboard.
- Delegasi (opsional)**
Manager dapat mendelegasikan approval ke Manager lain dengan scope waktu terbatas sebelum SLA habis

10 Entity Relationship Diagram (ERD)

📌 ERD mencerminkan schema PostgreSQL aktual. Data finansial detail proyek disimpan di `projects.detail` (JSONB). Tabel `organization_units` adalah entitas baru di v2.0 yang merepresentasikan 4 Pusat + 9 SBU.



Entitas Pendukung

Tabel

organization_units

```
assumptions_master /
assumptions_history
```

duration_presets

sla_config / sla_events

categories

system_config

Fungsi

Master hierarki organisasi: 4 Pusat + 9 SBU, relasi parent-child

WACC, inflasi, threshold BCR global + riwayat perubahan

Preset durasi & ambang BCR per kategori

Konfigurasi SLA hari kerja, reminder, eskalasi per role

Master code CAPEX/OPEX

Konfigurasi sistem (maintenance mode, feature flags, dll.)

audit_logs

Jejak audit immutable semua aksi pengguna

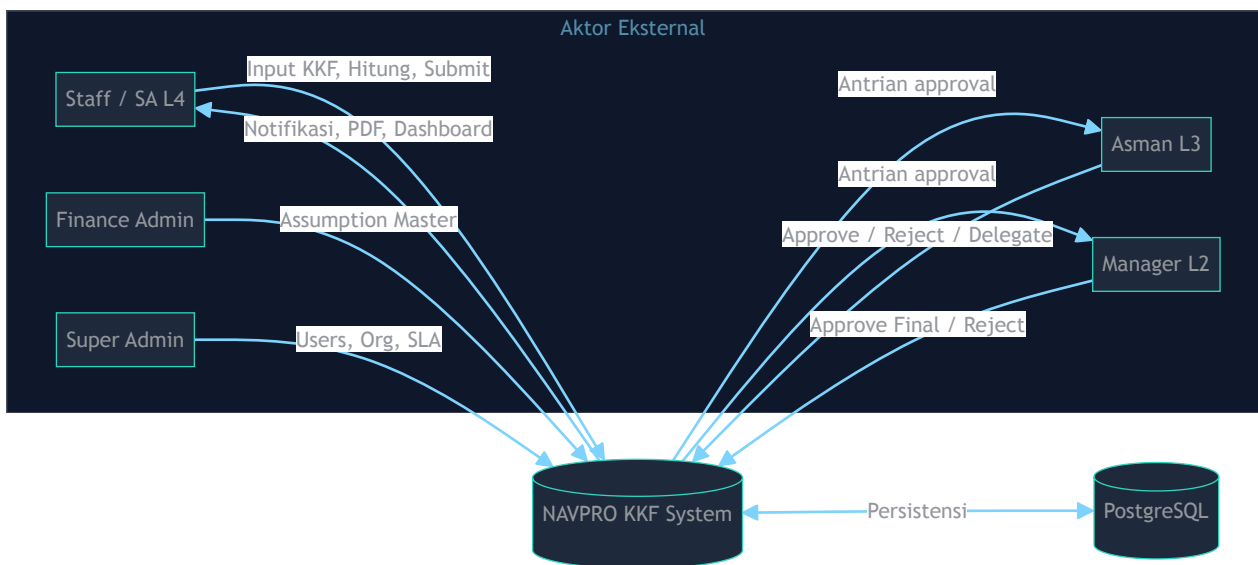
notifications

In-app notification matrix

11 Data Flow Diagram (DFD)

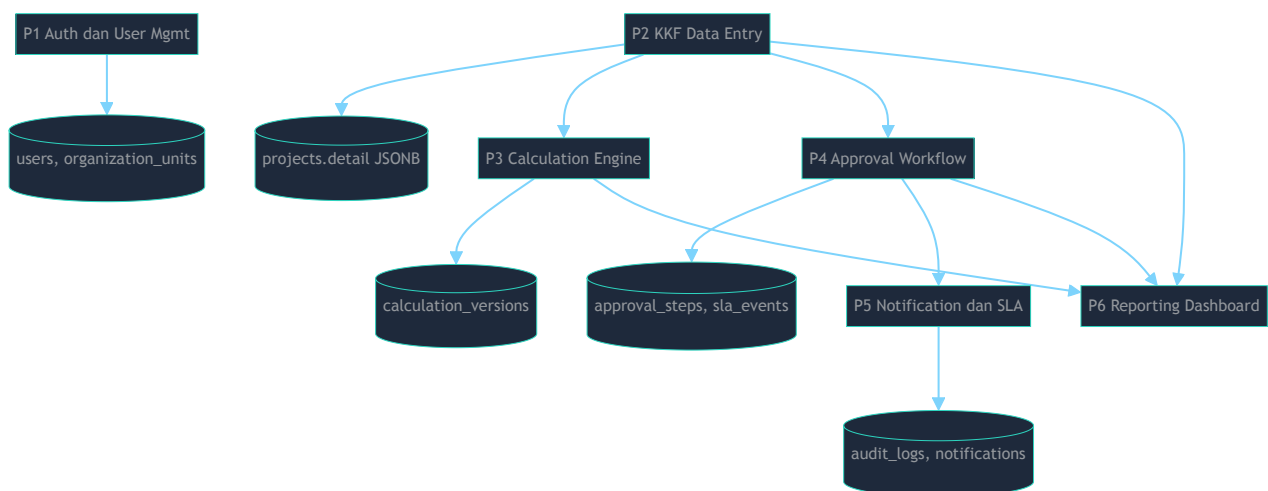
11.1 DFD Level 0 — Context Diagram

Menggambarkan aliran data antara seluruh aktor eksternal dengan sistem NAVPRO KKF dan database PostgreSQL.

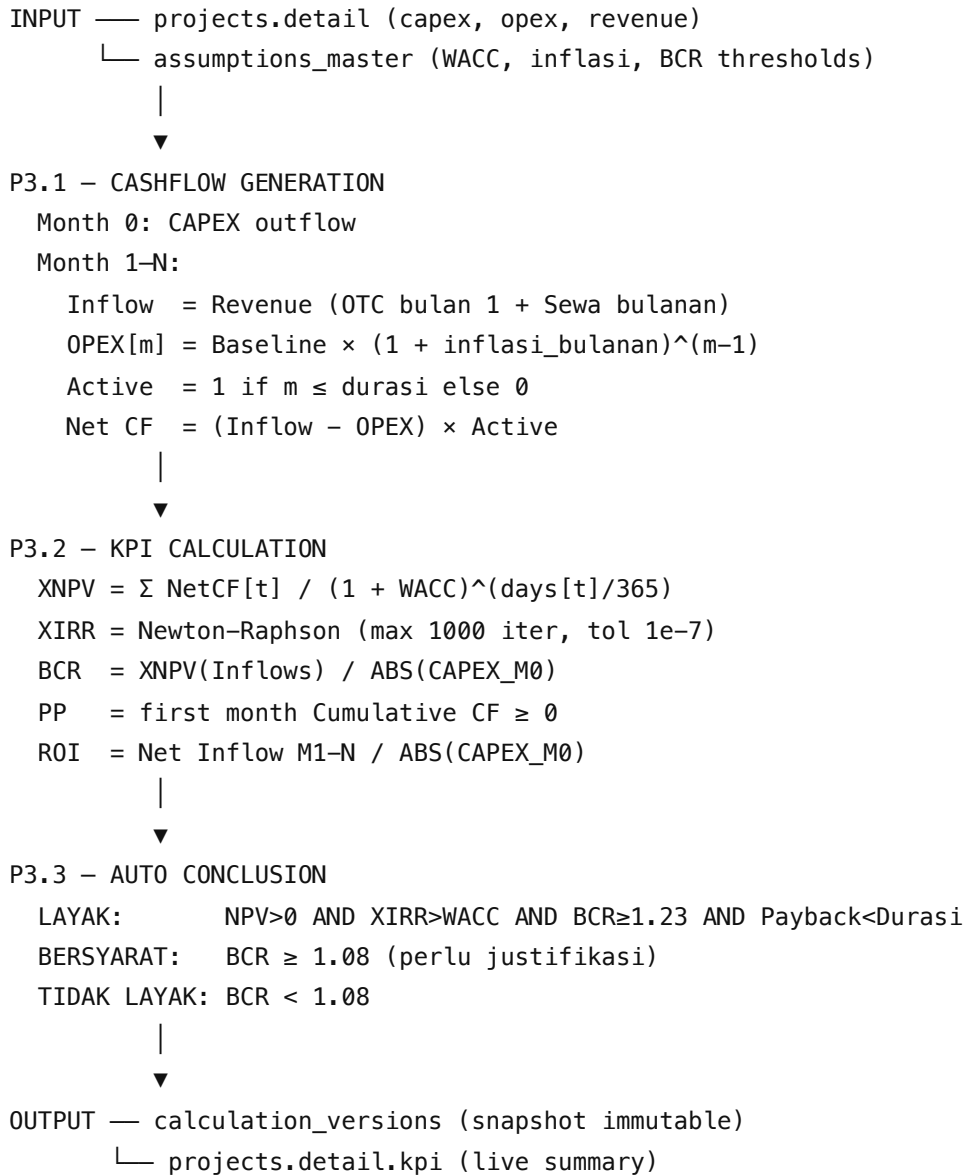


11.2 DFD Level 1 — Proses Utama

Enam proses utama sistem dan relasi aliran data ke store database masing-masing.



11.3 DFD Level 2 — Calculation Engine (P3)



12 Database Schema & Relasi

12.1 Kardinalitas Relasi

Parent	Child	Kardinalitas	FK	On Delete
organization_units	organization_units	1:N	parent_id	—
organization_units	users	1:N	org_unit_id	—
organization_units	projects	1:N	org_unit_id	—
users	projects	1:N	created_by	SET NULL
projects	calculation_versions	1:N	project_id	CASCADE
projects	approval_steps	1:N	project_id	CASCADE

projects	sla_events	1:N	project_id	CASCADE
approval_steps	sla_events	1:N	approval_step_id	CASCADE
users	approval_steps	1:N	assigned_to	SET NULL
users	notifications	1:N	user_id	CASCADE

12.2 Isolasi Data (RLS + API Scope)

Role	Scope Query projects
SUPER_ADMIN , FINANCE_ADMIN	Semua proyek
VP_SA	Semua segment
MANAGER	segment = segment user
ASMAN	org_unit_id = user.org_unit_id
STAFF	created_by = user.id

12.3 Status Proyek (State Reference)

Status	Deskripsi
DRAFT	Input / revisi data — dapat diedit oleh Staff
COMPUTED	Kalkulasi selesai, belum submit
SUBMITTED	Tersubmit, data terkunci, routing ke Asman
IN_REVIEW_ASMAN	Menunggu review Asman (step 1)
IN_REVIEW_MANAGER	Menunggu approve Manager (step 2)
APPROVED	Disetujui final oleh Manager
REJECTED	Ditolak → kembali ke DRAFT untuk revisi
ARCHIVED	Diarsipkan otomatis setelah 30 hari APPROVED

12.4 Schema SQL Utama

```
-- ORGANIZATION UNITS (tabel baru v2.0)
CREATE TABLE organization_units (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  code        VARCHAR(30) UNIQUE NOT NULL,
  name        VARCHAR(200) NOT NULL,
```

```

type          VARCHAR(20) NOT NULL CHECK (type IN ('PUSAT','SBU')),
segment       VARCHAR(20) CHECK (segment IN ('ENT1','ENT2','PLN1','PLN2')),
parent_id     UUID REFERENCES organization_units(id),
is_active     BOOLEAN DEFAULT true,
created_at    TIMESTAMPTZ DEFAULT now()
);

-- USERS (v2.0: tambah org_unit_id, org_level)
CREATE TABLE users (
  id           UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  email        VARCHAR(255) UNIQUE NOT NULL,
  password_hash VARCHAR(255) NOT NULL,
  full_name    VARCHAR(200) NOT NULL,
  employee_id  VARCHAR(50),
  role         VARCHAR(30) NOT NULL CHECK (role IN (
    'SUPER_ADMIN','FINANCE_ADMIN','VP_SA',
    'MANAGER','ASMAN','STAFF','VIEWER')),
  org_unit_id  UUID REFERENCES organization_units(id),
  org_level    VARCHAR(5) CHECK (org_level IN ('L1','L2','L3','L4')),
  is_active    BOOLEAN DEFAULT true,
  failed_login_count INTEGER DEFAULT 0,
  locked_until TIMESTAMPTZ,
  last_login_at TIMESTAMPTZ,
  created_at   TIMESTAMPTZ DEFAULT now()
);

-- PROJECTS (v2.0: tambah org_unit_id, segment)
CREATE TABLE projects (
  id           UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  project_code VARCHAR(50) UNIQUE NOT NULL,
  project_name VARCHAR(300) NOT NULL,
  created_by   UUID NOT NULL REFERENCES users(id),
  org_unit_id  UUID NOT NULL REFERENCES organization_units(id),
  segment      VARCHAR(20),
  status       VARCHAR(30) NOT NULL DEFAULT 'DRAFT',
  customer_name VARCHAR(300),
  project_duration_months INTEGER CHECK (project_duration_months BETWEEN 1 AND 120),
  detail       JSONB,
  current_version INTEGER DEFAULT 0,
  submitted_at TIMESTAMPTZ,
  approved_by  UUID REFERENCES users(id),
  approved_at  TIMESTAMPTZ,
  created_at   TIMESTAMPTZ DEFAULT now(),
  updated_at   TIMESTAMPTZ DEFAULT now()
);

-- APPROVAL STEPS (2 level: ASMAN=step 1, MANAGER=step 2)
CREATE TABLE approval_steps (
  id           UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  project_id   UUID NOT NULL REFERENCES projects(id) ON DELETE CASCADE,
  step_order   INTEGER NOT NULL,

```

```

    approver_level VARCHAR(20) NOT NULL CHECK (approver_level IN ('ASMAN','MANAGER')),
    assigned_to     UUID REFERENCES users(id),
    org_unit_id     UUID REFERENCES organization_units(id),
    status          VARCHAR(20) NOT NULL DEFAULT 'PENDING',
    comments        TEXT,
    due_at          TIMESTAMPTZ,
    acted_at        TIMESTAMPTZ,
    is_escalated    BOOLEAN DEFAULT false,
    created_at      TIMESTAMPTZ DEFAULT now(),
    UNIQUE(project_id, step_order)
);

-- SLA CONFIG SEED
INSERT INTO sla_config (role_key, role_name, sla_working_days, reminder_hours, escalation_ho
VALUES
    ('ASMAN', 'Assistant Manager SA', 2, 24, 48, 'MANAGER'),
    ('MANAGER', 'Manager SA Segment', 1, 12, 24, 'VP_SA');

```

12.5 Migration v1.0 → v2.0

```

-- 1. Buat tabel baru organization_units
-- (lihat CREATE TABLE di atas)

-- 2. Migrasi kolom users
ALTER TABLE users ADD COLUMN org_unit_id UUID REFERENCES organization_units(id);
ALTER TABLE users ADD COLUMN org_level   VARCHAR(5);
ALTER TABLE users ADD COLUMN employee_id VARCHAR(50);

-- 3. Migrasi kolom projects
ALTER TABLE projects ADD COLUMN org_unit_id     UUID REFERENCES organization_units(id);
ALTER TABLE projects ADD COLUMN segment         VARCHAR(20);
ALTER TABLE projects ADD COLUMN rejection_reason TEXT;

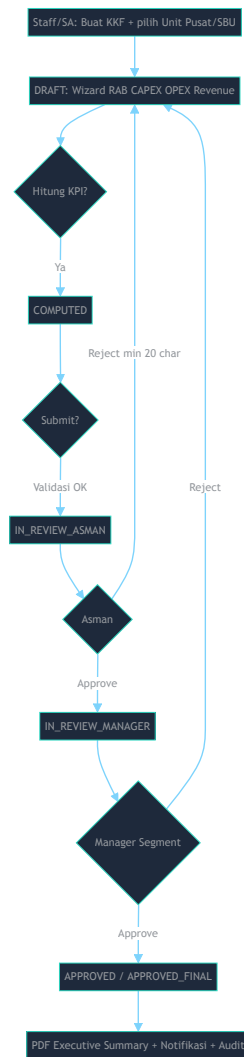
-- 4. Update SLA config: role lama → role baru
UPDATE sla_config SET role_key = 'ASMAN'   WHERE role_key = 'manager';
UPDATE sla_config SET role_key = 'MANAGER' WHERE role_key = 'gm_srm';

```

13 Workflow Business Process & State Machine

13.1 Business Process End-to-End

Alur bisnis lengkap dari inisiasi KKF oleh Staff hingga approval final oleh Manager Segment.



13.2 State Machine Approval

Diagram status proyek beserta semua transisi valid dan aktor yang bertanggung jawab.



13.3 Approval Steps Detail

step_order	approver_level	Routing	SLA Default
1	ASMAN	User role ASMAN pada projects.org_unit_id yang sama	2 hari kerja
2	MANAGER	Manager pada segment yang sama dengan unit proyek	1 hari kerja



Setelah submit: calculation_versions snapshot dibuat, data proyek terkunci, sla_events untuk reminder/eskalasi dibuat, dan notifikasi matrix dikirimkan (SUBMITTED, APPROVED, REJECTED, REMINDER, ESCALATION).

13.4 Transisi Status Valid

Dari	Ke	Trigger	Kondisi
DRAFT	SUBMITTED	Staff submit	Semua field mandatory + KPI sudah dihitung
SUBMITTED	IN_REVIEW_ASMAN	System auto	Setelah submit berhasil divalidasi
IN_REVIEW_ASMAN	IN_REVIEW_MANAGER	Asman approve	Asman pada org_unit yang sama dengan proyek
IN_REVIEW_ASMAN	DRAFT	Asman reject	Komentar ≥20 karakter wajib; notifikasi ke Staff
IN_REVIEW_MANAGER	APPROVED	Manager approve	Manager pada segment yang sama
IN_REVIEW_MANAGER	DRAFT	Manager reject	Komentar ≥20 karakter wajib; notifikasi ke Staff + Asman
APPROVED	ARCHIVED	System auto	30 hari setelah APPROVED

14 Notification Matrix

Event	Penerima	Channel	Tipe
KKF di-submit	Asman terkait (org_unit)	Email + In-App	APPROVAL
Asman approve → step 2	Manager segment	Email + In-App	APPROVAL
Asman reject	Staff pemilik	Email + In-App	REJECTION
Manager approve (final)	Staff + Asman + VP_SA	Email + In-App	INFO
Manager reject	Staff + Asman	Email + In-App	REJECTION
SLA T-1 hari (Asman)	Asman aktif	Email + In-App	REMINDER
SLA T-12 jam (Manager)	Manager aktif	Email + In-App	REMINDER
SLA terlampaui — eskalasi	Approver + Level atas	Email + In-App	ESCALATION
WACC/BCR diupdate	Semua Staff dengan DRAFT aktif	In-App	INFO
User baru dibuat	User baru (email)	Email	INFO

15

SLA Configuration

Role	SLA Working Days	Reminder	Eskalasi ke	Jam Kerja
ASMAN (L3)	2 hari kerja	T-1 hari (24 jam)	MANAGER (L2)	Sen–Jum, 08.00–17.00 WIB
MANAGER (L2)	1 hari kerja	T-12 jam	VP_SA (L1)	Sen–Jum, 08.00–17.00 WIB

17

Hari kerja: Senin–Jumat, tidak termasuk hari libur nasional Indonesia. SLA countdown dimulai saat status berubah ke IN_REVIEW_ASMAN atau IN_REVIEW_MANAGER.

16

Data Governance & Retention

Domain Data	Pemilik	Rule Governance	Retensi
Project Identity & Data	Staff/SA	Immutable setelah SUBMITTED	Indefinite
Financial Assumptions	Finance Admin	SA tidak dapat memodifikasi; setiap perubahan wajib referensi sumber	7 tahun
Approval Decisions	Asman/Manager	Immutable setelah dicatat	7 tahun
Audit Log	System	Tidak ada user yang dapat menghapus; append-only	5 tahun minimum
KKF Approved Archives	System	Accessible selama retention period	7 tahun
KKF Rejected/Cancelled	System	—	3 tahun
Notification Records	System	Auto-purge setelah retention	90 hari
Login/Security Logs	System	—	2 tahun

17

UI/UX Guidelines

17.1 Filosofi Desain

Tiga persona utama: (1) **Staff/SA** — deep multi-step data entry; (2) **Asman/Manager** — rapid approval action; (3) **Eksekutif** — at-a-glance portfolio intelligence. Prinsip: *"Executive readability at the top,*

operational depth on drill-down."

17.2 Halaman Utama

- **Login:** Clean centered card, ICON+/PLN logo, email+password, MFA sebagai secondary step
- **Dashboard Eksekutif (VP/Manager):** 4 KPI cards, bar chart monthly KKF, donut chart status, approval pipeline gauge, top 10 CAPEX table
- **Dashboard Operasional (Staff/Asman):** My projects list + status badges, pending approval queue, SLA alert banner
- **Wizard KKF:** Left-side stepper 7 langkah, progress indicator, right-side KPI preview panel auto-update, auto-save draft
- **Approval Review:** Read-only summary, KPI cards color-coded, cashflow table expandable, audit history timeline, Approve/Reject modal

17.3 Visual Design

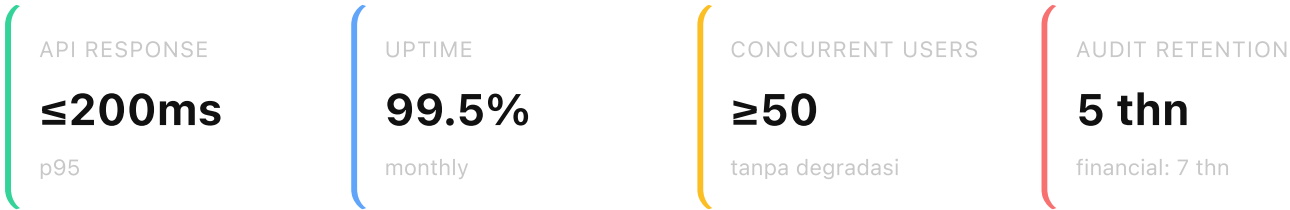
- **Warna:** Navy #1E3A5F (primary), Teal #2E7D7B (accent), Gold #C9A961 (highlight)
- **Tipografi:** Inter/Noto Sans; 14px body, 16px heading, 12px tabel
- **Status badges:** Draft (grey), Submitted (blue), In Review (orange), Approved (green), Rejected (red)
- **Format angka:** Locale ID — titik sebagai pemisah ribuan, koma sebagai desimal (Rp 1.250.000,00)
- **Mobile:** Approval action sebagai floating action button
- **Loading states:** Skeleton screens untuk halaman data-heavy

18 KPI & Success Metrics

18.1 Business KPIs

KPI	Baseline (AS-IS)	Target (TO-BE)	Improvement
Waktu proses KKF	3–5 hari kerja	<4 jam	~90%
Waktu approval	2–5 hari (kertas)	<24 jam (digital)	~80%
Konsistensi asumsi	~60%	100% (system-enforced)	+40%
Error formula	Tidak terukur	0% (server-side engine)	100%
Audit trail coverage	0%	100%	Full
Visibility eksekutif	Manual, delay hari	Real-time	Instant

18.2 System KPIs



19

Risiko & Mitigasi

ID	Risiko	Prob	Impact	Mitigasi
R-01	Staff salah assign org unit saat buat KKF	Medium	High	Dropdown org unit terikat ke login user, tidak dapat pilih unit lain
R-02	Asman tidak tersedia (cuti/kosong)	Medium	High	Delegation mechanism + eskalasi otomatis ke Manager saat SLA habis
R-03	SLA terlampaui di semua level	Low	High	Eskalasi otomatis ke VP_SA + notifikasi + dashboard SLA breach
R-04	Inkonsistensi data antar versi kalkulasi	Low	Critical	Snapshot immutable per submit; data locked setelah submit
R-05	Akses data lintas segment (breach)	Low	Critical	RLS PostgreSQL + RBAC di API layer; unit test per scope query
R-06	Perubahan WACC mempengaruhi KKF submitted	Medium	High	WACC snapshot tersimpan per calculation_version; tidak berubah retroaktif
R-07	Performance degradasi pada banyak proyek concurrent	Medium	Medium	Database indexing, connection pooling, PM2 cluster mode
R-08	Kehilangan data saat server restart	Low	Critical	PM2 auto-restart + PostgreSQL WAL + daily automated backup
R-09	Formula kalkulasi tidak paritas dengan Excel	Medium	Critical	Golden test 1 proyek dari KKF_Proyek_Investasi_1Tahun_v01.xlsx; unit test coverage ≥70%

20

Roadmap & Scope

20.1 In Scope (Phase 1)

- KKF project lifecycle: DRAFT → APPROVED → ARCHIVED
- 7 modul entry wizard (sesuai sheet Excel v01)

- Server-side calculation engine: XIRR, XNPV, BCR/PI, Payback, Simple ROI
- Assumption Master + riwayat perubahan
- Multi-level approval workflow: Asman → Manager, SLA, eskalasi
- Struktur organisasi: 4 Pusat + 9 SBU, RBAC per unit/segment
- Executive & operational dashboard
- Audit trail lengkap
- Export PDF Executive Summary + Excel cashflow
- In-app & email notifications
- Durasi proyek fleksibel: 1–120 bulan

20.2 Out of Scope (Phase 1)

- Integrasi SAP ERP real-time (Phase 2)
- Integrasi iCRM / LKPP (Phase 2)
- Mobile native app (iOS/Android)
- Retrieval WACC otomatis dari sistem perbankan
- AI-assisted feasibility scoring (Phase 3)
- Customer-facing portal
- Multi-currency calculation engine
- SCADA/OT integration
- Automated billing/invoicing dari KKF approved

20.3 Roadmap Fase Delivery

Batch	Fase	Konten	Gate
A — Wajib	0 → 1 → 2	Auth, wizard KKF, calculation engine, approval workflow ASMAN+MANAGER	Core KKF operasional harian
B — Laporan	3	Export PDF/Excel, Executive Summary, Version history UI	Pengganti Excel untuk presentasi
C — Operasi IT	4	CMS Admin: Assumption, SLA, Org Unit, User Management	Parameter tanpa deploy ulang
D — Production	5	Docker compose, test coverage, security pass, E2E Playwright	Staging UAT sign-off



Dokumen ini adalah **single source of truth** BRD NAVPRO KKF v2.1 — menggabungkan BRD KKF Proyek Investasi v1.0 (ICON+) dengan revisi arsitektur organisasi dan approval workflow NAVPRO v2.0. Perubahan selanjutnya mengikuti update kebijakan DirKeu atau struktur organisasi Bidang Solution Architect.